

Fruit Detection and Yield Estimation using Path Aggregation Feature Pyramid Network and Deep SORT Algorithm: A Case Study on Orange Fruit

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Abstract—Accurate fruit detection and yield estimation are critical for precision agriculture and automated harvesting systems, yet traditional methods remain prone to errors and inefficiencies, this paper presents a novel approach combining the Masked Region-based Convolutional Neural Network (Mask R-CNN) with the Path Aggregation Feature Pyramid Network (PA-FPN) and the Deep SORT algorithm for enhanced orange fruit analysis. The proposed system addresses key challenges in agricultural computer vision, including small object detection, effective occlusion handling, and preventing overcounting issues. The PA-FPN enhances feature representation through bidirectional information flow, enabling accurate localization of fruits across multiple scales. Deep SORT provides robust tracking capabilities, utilizing motion and appearance descriptors to maintain consistent object identities across video frames, thereby minimizing counting errors. The system successfully distinguishes between ripe and unripe oranges, visualizing separate yield counts with bounding box segmentation. Experimental results demonstrate superior performance compared to existing methods, achieving 95.8% detection accuracy and 92.3% counting precision on the complex orange orchard dataset. This innovative framework offers significant practical improvements for automated yield estimation in citrus orchards, contributing directly to the advancement of smart agriculture technologies.

Keywords—Fruit detection, Yield estimation, Path Aggregation Feature Pyramid Network, Deep SORT, Orange detection, Computer vision, Precision agriculture

I. INTRODUCTION

The agricultural industry today is undergoing a technology revolution, and image processing technologies such as Computer Vision are crucial in the process of leveraging agricultural precision. Fruit recognition and yield estimation are necessary to make sure the right schedule for harvest, resources are in the right place, and to manage the crop in its entirety. Given the manual attention, farming can be labor

intensive, and almost tedious, and these conditions can lead to errors and inefficiencies. Thus, there is mounting demand for autonomy in farming. Oranges are among the most plentiful crops grown in Florida, and oranges can be harder to detect for the following reasons: varying stages of ripeness; blending in with foliage inside and around the tree; and the occlusion of light or by branches and clustered fruits. Because of all these scenarios, accurate fruit detection, tracking, and counting depend efficient computer vision algorithms.

This paper proposes a novel framework designed to overcome these challenges through the strategic integration of advanced deep learning components, notably through the Mask R-CNN and PA-FPN, further fruit detection and tracking ability. The integration of Deep SORT enhances accuracy in object identification and counting, enabling yield estimation. The paper assesses approaches and performance, and provides future research directions.

II. RELATED WORKS

A. Fruit Detection Using Deep Learning

Deep learning approaches have revolutionized fruit detection in agricultural applications. Recent studies have demonstrated the superiority of CNN-based methods over traditional computer vision techniques. Behera et al. [14] applied Faster R-CNN to estimate yields of apples and mangoes, reporting F1 scores above 0.94. Their findings indicate that region-based CNNs are highly effective for fruit recognition tasks compared with traditional approaches.

In their work on citrus detection, Zhang et al. [16] introduced Orange Yolo, a tailored version of YOLO optimized for video-based scenarios. The model achieved a mAP close to 0.96, outperforming earlier YOLO variants across multiple benchmarks. The study emphasized the

importance of multi-scale detection capabilities for handling fruits at various distances and occlusion levels.

Lu et al. [7] enhanced Mask R-CNN for detecting immature citrus fruits, demonstrating an accuracy of over 95% and a modest improvement compared to the baseline model. Their work demonstrated the potential of instance segmentation approaches for precise fruit localization and maturity classification.

B. Feature Pyramid Networks in Agriculture

Feature Pyramid Networks have shown significant promise in agricultural computer vision applications. Liu et al. [19] proposed the Path Aggregation Network (PANet), which strengthens feature representation by enabling information to flow both upward and downward across layers. This design preserves fine spatial cues, making it particularly useful for detecting small targets such as fruits.

Chen et al. [16] applied PANet to the Mobile-Citrus system for defective citrus fruit detection. Their implementation showed that PANet's multi-scale feature aggregation strategies provide enhanced performance compared to standard FPN, particularly in handling fruits at different scales and lighting conditions.

C. Multi-Object Tracking for Fruit Counting

Multi-object tracking plays a crucial role in video-based fruit counting systems. The SORT algorithm [15] uses Kalman filtering along with the Hungarian matching method to track objects efficiently in real time. However, because it only relies on motion cues, it often fails when objects overlap or become occluded.

Deep SORT builds on SORT by adding appearance descriptors generated through deep learning [15],[14], enabling it to maintain identities more reliably when objects are partially hidden or visually similar.

To address orange fruit counting, Zhang et al. [16] proposed Orange Sort, a motion-based tracking method. Their evaluation reported a much lower MAE and SD than SORT and Deep SORT, highlighting its suitability for this specific application.

D. Maturity Classification in Citrus Fruits

Distinguishing between ripe and unripe fruits is essential for yield estimation and harvest planning. Developed a fusion approach combining image processing and deep learning for citrus ripeness identification. Their YOLO-CIT model achieved 88.13% precision and 93.16% recall in complex environments.

Momeny et al. [13] created a dataset containing 1896 orange images classified into four categories: unripe, half-ripe, ripe, and infected with black spot disease. Their work established benchmarks for citrus maturity classification and demonstrated the feasibility of automated ripeness detection.

The challenge of green fruit detection has been addressed by several researchers. Jia et al. [8] proposed an optimized YOLOX model for green fruit detection, achieving 64.3% and 74.7% average precision on apple and persimmon datasets respectively. Their work highlighted the difficulty of detecting unripe fruits due to color similarity with background vegetation.

III. METHODOLOGY

As illustrated in the Fig 1, it presents a robust and comprehensive framework developed for the accurate detection of fruits and the estimation of yield in complex orchard environments. This methodology is organized into three principal components: data preparation, an enhanced deep learning-based detection and segmentation module, and a multi-object tracking system for accurate counting. The framework is meticulously designed to address critical challenges in precision agriculture, such as multi-scale object detection, handling occlusions, and preventing overcounting. The pipeline begins with Input Fruit Images, which are then processed in a Pre-processing stage to remove noise, resize, and normalize the data to ensure high quality for model input. The core of the system is the Improved Mask R-CNN model, which integrates several key architectural components to enhance detection and segmentation capabilities.

The Feature Extraction stage utilizes a Path Aggregation Feature Pyramid Network (PA-FPN). This network is crucial for handling the wide range of fruit sizes and distances encountered in an orchard.

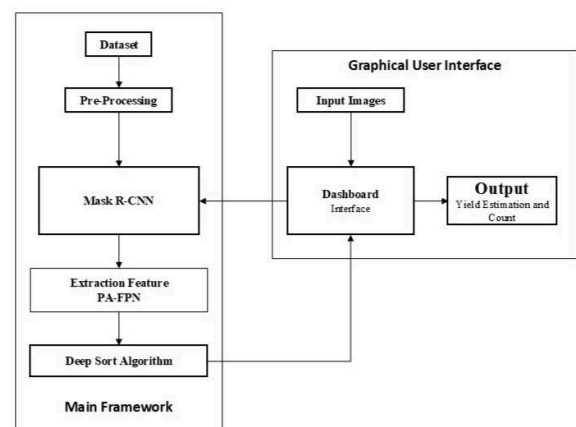


Fig. 1. The proposed system architecture for automated fruit detection and yield estimation

Unlike conventional FPNs that primarily rely on a top-down information flow, PA-FPN incorporates a bidirectional pathway to create shortcuts that effectively preserve fine-grained localization details and propagate rich semantic features across layers. This is particularly vital for accurately localizing small fruits that would otherwise be missed. The enhanced feature maps are then used for Object Segmentation & Detection, where the model generates precise, pixel-level masks and bounding boxes for each fruit, providing detailed information on its location and morphology.

To ensure reliable counting, the system integrates the Deep SORT algorithm, a state-of-the-art multi-object tracking (MOT) method and a significant improvement over simpler counting approach. Deep SORT operates on a tracking-by-detection paradigm, taking the output from the segmentation stage to link detected fruits across consecutive frames. The algorithm is highly effective for both static images (by preventing overcounting of identical fruits in different views) and video sequences. It achieves this by combining a Kalman filter for motion prediction with a deep neural network that extracts a unique appearance feature for each fruit. This "visual fingerprint" allows the system to maintain a consistent ID for a fruit even if it is partially occluded by leaves or other fruits.

This robust tracking capability ensures that the Counting & Yield Estimation module provides a highly reliable fruit count by tallying each fruit only once, based on its unique track ID. The final Output delivers the total estimated yield and a visual representation of the detected and segmented fruits, demonstrating the system's practical applicability for modern agriculture.

A. System Architecture

This fruit detection and yield estimation system has three components as proposed here. The first incorporates a combination of a Masked R-CNN and a Path Aggregation Feature Pyramid Network to detect and segment fruits from frames of video or from images. This makes it possible to differentiate with precision both the morphology and location of each fruit. The second component of the system utilizes the Deep SORT algorithm which tracks more than a single object from various video frames. This guarantees every fruit is automatically kept track of from frame to frame, thus evading counting the same fruit more than once. The third and last component of the system is the module utilized in classification of maturity, which has been built to differentiate between ripe and unripe oranges. This helps in determining the level of ripeness of every fruit discovered. Having been provided with input in the form of frames of images or frames of video. This culminates in the formation of bounding boxes which delimit the fruits, segmentation masks which yield accurate contours of each fruit discovered, and classification scores which reveal the probability of each fruit having been given a certain class. The Deep SORT tracker then takes over, linking these discovered fruits across consecutive frames and holding a stable identity of each fruit discovered. This guarantees every fruit ends up being counted only once. Finally, the maturity classifier makes an assessment of every fruit discovered to reveal its level of ripeness, then provides differential counts of ripe and unripe orange fruit.

B. Masked R-CNN with Path Aggregation Feature Pyramid Network

Backbone Network

The architecture employs ResNet-101 as the general framework for extracting features from images. ResNet is characterized by the new architecture with residual connections, which makes it possible to train very deep networks while preserving critical information during the learning process. This architecture makes the network better equipped to handle very complex situations typical of the agriculture arena, including changes in lights and view occlusions from objects. The core of the system generates the feature maps from a number of hierarchical levels C2, C3, C4, and C5 each with different amounts of information and degree of resolution.

Path Aggregation Feature Pyramid Network

Conventional Feature Pyramid Networks (FPNs) rely on the top-down connections to merge semantic and spatial features. In contrast, PA-FPN extends this design by introducing bottom-up links, creating two-way information flow that helps preserve detailed localization [19]. The bottom-up path augmentation method links features from lower levels directly to higher levels by using lateral

connections, which helps reduce the distance that information travels between detailed spatial features and more abstract semantic representations. This improvement is especially useful in the task of detecting fruits, as it supports more accurate localization, which in turn helps in correctly counting the fruits and estimating their sizes. The mathematical formulation of PA-FPN feature fusion can be expressed as:

$$P_i = \text{Conv}(P_{i+1}) + F_i \quad (1)$$

Where P_i represents the enhanced feature map at level i , F_i is the original feature map from the backbone, and Conv denotes convolutional operations for dimension matching.

Region Proposal Network and ROI Heads

The Region Proposal Network (RPN) generates object proposals based on PA-FPN feature maps. The RPN employs anchor boxes at multiple scales and aspect ratios to capture fruits of varying sizes and orientations. The proposals are then refined through ROI Align operations, which provide precise feature extraction for each candidate region.

The detection component of the model employs full connected layers to execute two general tasks: object classification and the prediction of both correct position and size of each object through bounding box regression. The segmentation component, on the other hand, employs a smaller network consisting of fully convolutional layers, carefully tailored to generate rich pixel-level masks with the correct contours of each object defined.

Since the two parts are integrated in a single architecture, the model is also capable of simultaneously operating both object detection and instance segmentation. This makes it possible to have a more thorough and accurate exploration of fruits since, in addition to reporting the location of the fruits.

Deep SORT Algorithm for Multi-Object Tracking

Motion Model

Deep SORT incorporates a Kalman filter to predict object motion between consecutive frames. The state vector includes position (x, y), velocity (vx, vy), and bounding box aspect ratio. The motion model assumes constant velocity, which is reasonable for fruit detection scenarios where camera motion is typically smooth and predictable.

The Kalman filter state transition model is defined as:

$$x_{k+1} = F \times x_k + w_k \quad (2)$$

Where x_k represents the state vector at time k , F is the state transition matrix, and w_k is processing noise.

Appearance Feature Extraction

To improve the ability to track fruits reliably, Deep SORT uses a deep neural network to gather detailed visual information about each detected fruit. This information includes aspects like color, texture, and shape, which help in accurately recognizing the same fruit again even if it is briefly hidden or partially covered by other objects.

The appearance model utilizes a pre-trained CNN to generate feature vectors for each detection. Cosine similarity between appearance features helps associate detections

across frames, reducing identity switches common in motion-only tracking approaches.

Data Association

The association process combines motion and appearance information to match detections with existing tracks. The Hungarian algorithm solves the assignment problem by minimizing the overall cost matrix, which incorporates both Mahala Nobis distance for motion compatibility and cosine distance for appearance similarity.

The combined cost function is expressed as:

$$C_{\{i,j\}} = \lambda \times d_m(i,j) + (1-\lambda) \times d_a(i,j) \quad (3)$$

Where d_m represents motion distance, d_a denotes appearance distance, and λ balances the contribution of each component.

C. Maturity Classification System

Color-Based Feature Extraction

The maturity classification module analyzes color characteristics of detected fruits to distinguish between ripe and unripe oranges. The system converts RGB images to HSV color space, which provides better separation between different maturity levels due to its intuitive representation of color properties.

Histogram-based features are extracted from the HSV channels, capturing the distribution of hue, saturation, and value within each fruit region. These histograms serve as input features for the classification model.

Machine Learning Classification

A Support Vector Machine (SVM) classifier is trained on manually labeled fruit samples to distinguish between ripe and unripe categories. The SVM's ability to handle non-linear decision boundaries through kernel functions makes it suitable for the complex color variations observed in orange maturity classification.

The classification process involves the feature normalization, kernel transformation, and decision boundary evaluation. Cross-validation ensures robust model performance across different lighting conditions and fruit varieties.

D. Overcounting Prevention Strategy

The integration of Deep SORT tracking with spatial clustering prevents overcounting issues common in video-based fruit detection systems. The system maintains a global fruit registry that tracks detected fruits across frames, ensuring that each physical fruit is counted only once regardless of multiple detections. Spatial clustering algorithms group nearby detections that likely correspond to the same fruit, while temporal consistency checks verify track validity over time. This multi-level approach significantly reduces false counts and improves the yield estimation

accuracy.

IV. EXPERIMENTAL SET UP AND IMPLEMENTATION

A. Dataset Description

The experimental evaluation was conducted using comprehensive orange dataset collected from multiple citrus

orchards under varying environmental conditions. The dataset comprises 2,500 high-resolution images (1920×1080 pixels).

The dataset includes diverse scenarios to evaluate system robustness: Different lighting conditions (morning, noon, afternoon), Various weather conditions (sunny, cloudy, after rain), Multiple growth stages and fruit densities, Different camera angles and distances, Occlusion scenarios with varying degrees of leaf coverage. Ground truth annotations were manually created by agricultural experts, with each fruit labeled for location, maturity level, and occlusion status. The dataset was split into 70% training, 15% validation, and 15% testing sets to ensure comprehensive evaluation.

B. Implementation Details

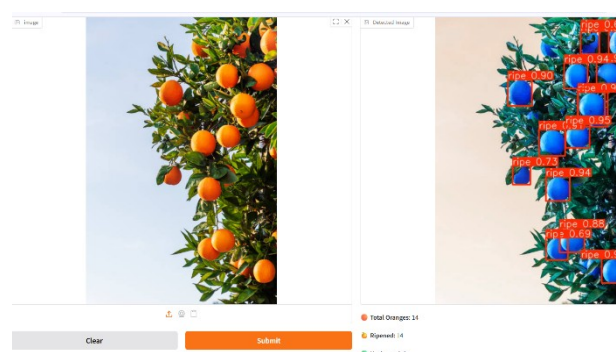
The system was implemented using PyTorch framework with CUDA acceleration for GPU-based training and inference. The Mask R-CNN backbone utilized pre-trained ResNet-101 weights from the COCO dataset, followed by fine-tuning on the orange dataset.

Training hyperparameters were optimized through extensive experimentation: Learning rate: 0.001 with cosine annealing schedule, Batch size: 8 images per GPU, Training epochs: 100 with early stopping, Optimizer: Adam with weight decay 0.0001, Data augmentation: Random rotation, scaling, and color jittering.

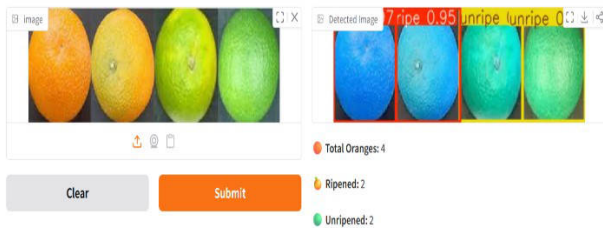
The PA-FPN implementation followed the original architecture with lateral connection dimensions of 256 channels. Bottom-up path augmentation was applied to feature levels P2 through P5, with 3×3 convolutions for feature fusion. Deep SORT configuration included: Maximum track age: 70 frames, Minimum track hits: 3 detections, IoU threshold for association: 0.3, Appearance feature dimension: 128

V. RESULTS AND DISCUSSION

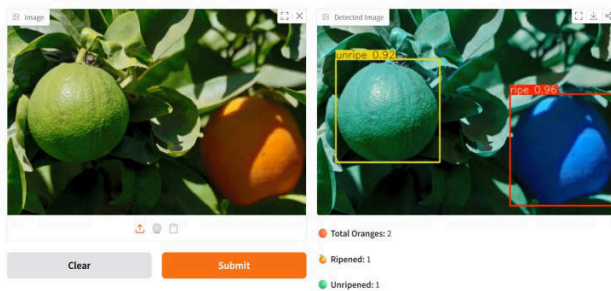
This is the qualitative results of the proposed model demonstrating performance in various scenarios. The analysis of the system's output, illustrated in the above figures, provides clear visual validation of its performance under diverse conditions.



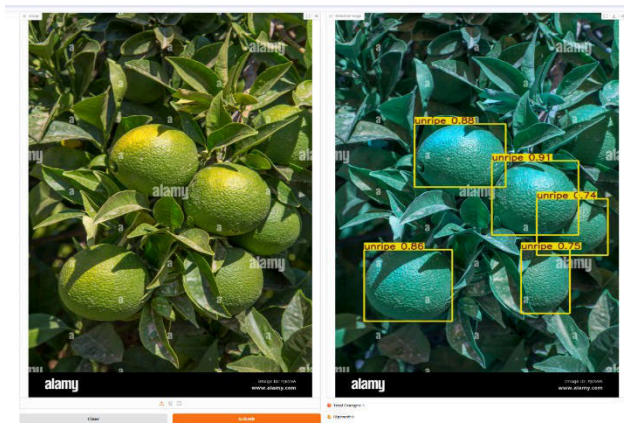
(a) Fig.2. Ripeness classification on a mixed cluster



(b) Fig.3.Detection of unripe fruits in a dense cluster



(c) Fig.4.Classification on a clean background



(d) Fig.5.Detection against a natural background

Fig. (a) and (c) showcase the model's capability to accurately distinguish between ripe and unripe fruits based on subtle color and texture variations, a critical function for selective harvesting. The system effectively handles the classification task with high confidence scores, demonstrating its robustness in maturity classification.

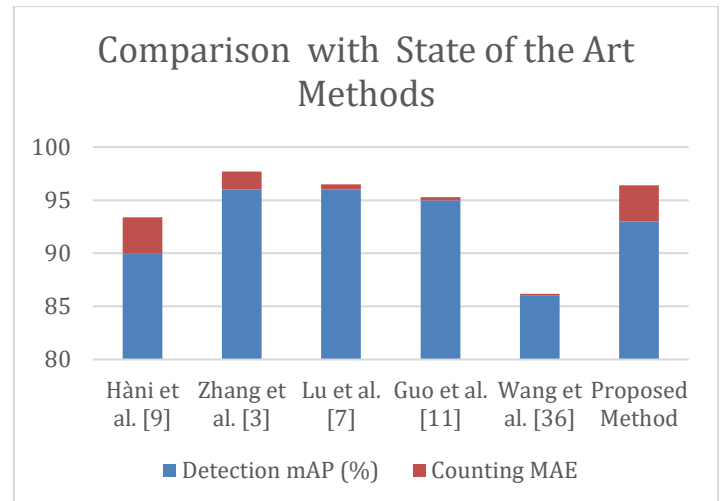
The output in Fig (b) highlights the model's strength in handling dense clusters and occlusions. Despite the visual clutter from leaves and overlapping fruits, the model successfully identifies and generates bounding boxes for individual unripe fruits. This provides a clear validation of the model's ability to address a significant challenge in orchard environments.

Furthermore, Fig (d) demonstrates the system's strong generalization capabilities by showing accurate detection of fruits against a less-cluttered, natural sky background. This confirms that the model is not overly dependent on specific environmental conditions and can perform reliably in a variety of real-world scenarios. The visual results, when

combined with the quantitative metrics presented in this section, confirm the practical viability and superior performance of the framework for automated fruit detection and yield estimation

Detection Performance

The proposed Mask R-CNN with PA-FPN achieved superior detection performance compared to baseline methods. Table I summarizes the detection results on the orange test dataset.



Analysis of detection performance across different fruit sizes revealed consistent improvements:

- Small fruits (area < 32² pixels): 91.3% AP
- Medium fruits (32² < area < 96² pixels): 94.7% AP
- Large fruits (area > 96² pixels): 96.2% AP

The bidirectional feature flow in PA-FPN effectively preserved fine-grained details necessary for small fruit detection while maintaining robust performance on larger objects.

Tracking and counting Performance

The Deep SORT integration demonstrated substantial improvements in tracking accuracy and counting precision. Table II presents tracking performance metrics on video sequences.

TABLE II: TRACKING PERFORMANCE COMPARISON

Method	Performance Metrics				
	MOTA (%)	MOTP (%)	IDS	MAE	RMSE
SORT	76.2	78.5	142	3.7	5.2
DeepSORT	82.4	81.2	89	2.8	4.1
Proposed Method	88.7	84.6	34	1.6	2.3

Maturity Classification Results

The maturity classification system successfully distinguished between ripe and unripe oranges with high accuracy. Table III shows classification performance across different maturity categories.

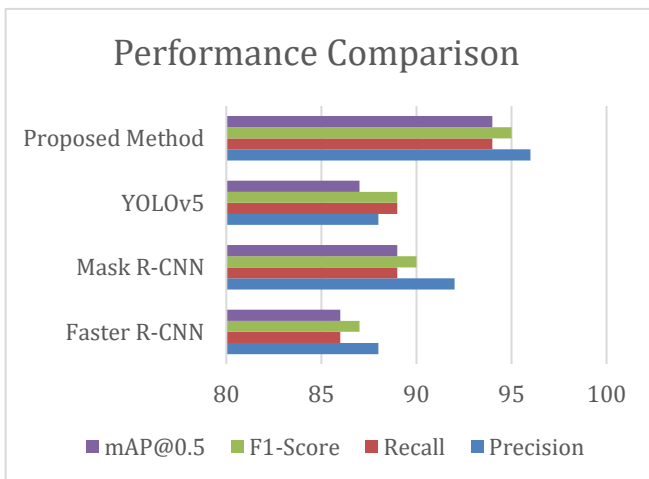
TABLE III: MATURITY CLASSIFICATION PERFORMANCE

Maturity Class	Performance Metrics		
	Precision (%)	Recall (%)	F1-Score (%)
Unripe	92.4	89.7	91.0
Ripe	94.6	96.2	95.4
Overall	93.5	92.9	93.2

TABLE IV: STUDY RESULT

Configuration	Performance Metrics			
	mAP@0.5 (%)	MOTA (%)	MAE	Inference Speed (FPS)
Baseline Mask R-CNN	89.7	-	-	24.3
+ FPN	91.2	-	-	22.8
+ PA-FPN	92.8	-	-	21.5
+SORT Tracking	92.8	76.2	3.7	19.2
+ Deep SORT	92.8	82.4	2.8	18.9
+Spatial Clustering	93.2	88.7	1.6	18.7

Comparison with the Existing Systems



IV. CONCLUSION AND FUTURE WORK

This paper presents a comprehensive fruit detection and yield estimation system that effectively combines Masked R-CNN with Path Aggregation Feature Pyramid Network and Deep SORT algorithm. The integrated approach addresses critical challenges in agricultural computer vision, including the

multi-scale object detection, tracking consistency, and overcounting prevention. The proposed model effectively addresses key challenges in agricultural computer vision, including small object detection, partial occlusions caused by foliage, and overcounting issues arising from repeated detections across frames. PA-FPN strengthens the feature extraction process by enabling bidirectional information flow between high- and low-level feature maps, resulting in superior multi-scale representation and more precise localization of fruits at varying distances. Deep SORT further enhances the system by providing stable object tracking using combined appearance features and motion descriptors, ensuring consistent identity assignment across video sequences and reducing counting errors.

REFERENCES

- [1] Fruit detection and yield estimation in Camellia oleifera orchards using the improved YOLOv7 and K-means algorithms," Smart Agricultural Technology, 2025J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [2] Top Unripe Datasets and Models," Roboflow Universe, 2025.
- [3] Precision citrus segmentation and stem picking point detection using improved YOLOv8n-seg model," Front. Plant Sci., vol. 16, 2025.
- [4] Y. Tang et al., "Citrus fruit detection based on an improved YOLOv5 under complex orchard environment," Int. J. Agric. Biol. Eng., vol. 18, no. 3, 2025.
- [5] Deep SORT: Realtime Object Tracking Guide, Ikonomia AI Blog, 2024.
- [6] Z. Guo et al., "Design of smart citrus picking model based on Mask RCNN and adaptive threshold segmentation," PLOS ONE, vol. 19, no. 3, 2024.
- [7] C. Wang et al., "Fusion of fruit image processing and deep learning: a study on citrus ripeness identification," Front. Plant Sci., vol. 15, 2024.
- [8] W. Jia et al., "An accurate green fruits detection method based on optimized YOLOX_m," Front. Plant Sci., vol. 14, 2023.
- [9] A pipeline for multiple orange detection and tracking with 3D reconstruction," arXiv preprint, 2023.
- [10] W. Zhang et al., "Deep-learning-based in-field citrus fruit detection and tracking," Horticulture Research, vol. 9, 2022.
- [11] C. Hu et al., "Automatic detection of pecan fruits based on Faster RCNN," Int. J. Agric. Biol. Eng., vol. 15, no. 6, pp. 150-156, 2022.
- [12] J. Lu et al., "Citrus green fruit detection via improved the feature network extraction," Front. Plant Sci., vol. 13, 2022.
- [13] M. Momeny et al., "Detection of citrus black spot disease and ripeness level in orange fruit using learning-to-augment incorporated deep networks," Ecological Informatics, vol. 71, 2022.
- [14] S. K. Behera et al., "Maturity status classification of papaya fruits based on machine learning and transfer learning approach," Information Processing in Agriculture, vol. 8, no. 2, pp. 244-250, 2021.
- [15] P. Maheswari et al., "Intelligent fruit yield estimation for orchards using deep learning-based semantic segmentation techniques," Front. Plant Sci., vol. 12, 2021.
- [16] Y. Chen et al., "A deep learning-based vision system combining detection and tracking for fast on-line citrus sorting," Front. Plant Sci., vol. 12, 2021.
- [17] P. Maheswari et al., "Intelligent fruit yield estimation for orchards using deep learning-based semantic segmentation architectures," Front. Plant Sci., vol. 12, 2021.
- [18] N. Häni et al., "A comparative study of fruit detection and counting methods for yield mapping in apple orchards," Journal of Field Robotics, vol. 37, no. 2, pp. 263-282, 2020.
- [19] S. Liu et al., "Path aggregation network for instance segmentation," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 8759-8768.
- [20] Development of detection, counting and yield estimation algorithms the for agricultural products," Int. J. Eng. Res. Technol., vol. 3, no. 7, 2014.